

HDOC DAY 18 : ASSIGNMENT

AIM: To write a menu-driven C program using switch-case and user-defined functions (with void return type and no arguments) to perform digit-based operations on a positive integer: checking if a number is a Strong, Armstrong, Harshad, or Duck number.

ALGORITHM:

1. Start

2. Display Menu:

- a. Option 1: Check Strong Number
- b. Option 2: Check Armstrong Number
- c. Option 3: Check Harshad Number
- d. Option 4: Check Duck Number
- e. Option 5: Exit

3. Read user choice.

4. Use switch-case to call the respective function based on choice:

a. If Choice 1 (checkStrong()):

- i. Read a positive integer num.
- ii. Store num in a temporary variable temp.
- iii. Initialize sum = 0.
- iv. Extract each digit using $\text{digit} = \text{temp} \% 10$.
- v. Calculate factorial of digit and add it to sum.
- vi. Remove last digit using $\text{temp} = \text{temp} / 10$.
- vii. Repeat until $\text{temp} == 0$.
- viii. If $\text{sum} == \text{num}$, print it is a Strong Number; else, print Not Strong.

b. If Choice 2 (checkArmstrong()):

- i. Read a positive integer num.
- ii. Store num in temp to find total digits count.

iii. Extract digits, compute $\text{sum} = \text{sum} + (\text{digit})^{\text{count}}$, and compare $\text{sum} == \text{num}$.

iv. Print whether it is an Armstrong Number or not.

c. If Choice 3 (checkHarshad()):

i. Read a positive integer num.

ii. Find the sum of all digits of num.

iii. If $\text{num} \% \text{sum} == 0$, print it is a Harshad Number; else, print Not Harshad.

d. If Choice 4 (checkDuck()):

i. Read a positive integer num.

ii. Check if any digit is 0 (excluding leading zeros).

iii. If 0 is found, print it is a Duck Number; else, print Not Duck.

5. Repeat menu using loop until the user selects Exit.

6. Stop

TEST CASE TABLE:

TEST CASE	OPTION SELECTED BY USER	INPUT NUMBER	EXPECTED OUTPUT	STATUS
1.	1	145	Strong number	Pass
2.	2	371	Armstrong number	pass
3.	3	18	Harshad number	pass
4.	4	205	Duck number	pass

C PROGRAM:

```
#include <stdio.h>
```

```
#include <math.h>
```

```
// Function Declarations
```

```
void checkStrong();
```

```
void checkArmstrong();
```

```
void checkHarshad();
```

```
void checkDuck();
```

```
int main() {
```

```
    int choice;
```

```
    do {
```

```
        printf("\n--- MENU ---");
```

```
        printf("\n1. Strong Number");
```

```
        printf("\n2. Armstrong Number");
```

```
        printf("\n3. Harshad Number");
```

```
        printf("\n4. Duck Number");
```

```
        printf("\n5. Exit");
```

```
        printf("\nEnter your choice: ");
```

```
        scanf("%d", &choice);
```

```
        switch (choice) {
```

```
            case 1:
```

```
                checkStrong();
```

```
        break;
    case 2:
        checkArmstrong();
        break;
    case 3:
        checkHarshad();
        break;
    case 4:
        checkDuck();
        break;
    case 5:
        printf("\nExiting program...\n");
        break;
    default:
        printf("\nInvalid choice! Please try again.\n");
    }
} while (choice != 5);

return 0;
}
```

// Function to check Strong Number

```
void checkStrong() {
```

```
int num, temp, digit, sum = 0, fact, i;

printf("\nEnter a positive integer: ");
scanf("%d", &num);

temp = num;
while (temp > 0) {
    digit = temp % 10;

    // Factorial calculation
    fact = 1;
    for (i = 1; i <= digit; i++) {
        fact = fact * i;
    }

    sum = sum + fact;
    temp = temp / 10;
}

if (sum == num) {
    printf("%d is a Strong Number.\n", num);
} else {
    printf("%d is NOT a Strong Number.\n", num);
}
```

```
}
```

```
// Function to check Armstrong Number
```

```
void checkArmstrong() {
```

```
    int num, temp, digit, count = 0, sum = 0;
```

```
    printf("\nEnter a positive integer: ");
```

```
    scanf("%d", &num);
```

```
    temp = num;
```

```
    // Count number of digits
```

```
    while (temp > 0) {
```

```
        count++;
```

```
        temp = temp / 10;
```

```
    }
```

```
    temp = num;
```

```
    while (temp > 0) {
```

```
        digit = temp % 10;
```

```
        sum = sum + pow(digit, count);
```

```
        temp = temp / 10;
```

```
    }
```

```
    if (sum == num) {
```

```
    printf("%d is an Armstrong Number.\n", num);  
} else {  
    printf("%d is NOT an Armstrong Number.\n", num);  
}  
}
```

// Function to check Harshad Number

```
void checkHarshad() {  
    int num, temp, digit, sum = 0;  
  
    printf("\nEnter a positive integer: ");  
    scanf("%d", &num);  
  
    temp = num;  
    while (temp > 0) {  
        digit = temp % 10;  
        sum = sum + digit;  
        temp = temp / 10;  
    }  
  
    if (num % sum == 0) {  
        printf("%d is a Harshad Number.\n", num);  
    } else {  
        printf("%d is NOT a Harshad Number.\n", num);  
    }  
}
```

```
}  
}
```

```
// Function to check Duck Number
```

```
void checkDuck() {
```

```
    int num, temp, digit, isDuck = 0;
```

```
    printf("\nEnter a positive integer: ");
```

```
    scanf("%d", &num);
```

```
    temp = num;
```

```
    while (temp > 0) {
```

```
        digit = temp % 10;
```

```
        if (digit == 0) {
```

```
            isDuck = 1;
```

```
            break;
```

```
        }
```

```
        temp = temp / 10;
```

```
    }
```

```
    if (isDuck == 1) {
```

```
        printf("%d is a Duck Number.\n", num);
```

```
    } else {
```

```
        printf("%d is NOT a Duck Number.\n", num);
```



```
}  
  
}
```

CODE SCREENSHOT :



The screenshot shows a C program in a code editor. The editor's title bar indicates 'UW PICO 5.09' and 'File: 18q1'. The code defines a menu-driven program that checks for Strong, Armstrong, Harshad, and Duck numbers. It includes a switch statement for the menu options and a function to check for Strong numbers.

```
UW PICO 5.09 File: 18q1  
  
#include <stdio.h>  
#include <math.h>  
  
// Function Declarations  
void checkStrong();  
void checkArmstrong();  
void checkHarshad();  
void checkDuck();  
  
int main() {  
    int choice;  
  
    do {  
        printf("\n--- MENU ---");  
        printf("\n1. Strong Number");  
        printf("\n2. Armstrong Number");  
        printf("\n3. Harshad Number");  
        printf("\n4. Duck Number");  
        printf("\n5. Exit");  
        printf("\nEnter your choice: ");  
        scanf("%d", &choice);  
  
        switch (choice) {  
            case 1:  
                checkStrong();  
                break;  
            case 2:  
                checkArmstrong();  
                break;  
            case 3:  
                checkHarshad();  
                break;  
            case 4:  
                checkDuck();  
                break;  
            case 5:  
                printf("\nExiting program...\n");  
                break;  
            default:  
                printf("\nInvalid choice! Please try again.\n");  
        }  
    } while (choice != 5);  
  
    return 0;  
}  
  
// Function to check Strong Number  
void checkStrong() {  
    int num, temp, digit, sum = 0, fact, i;  
  
    printf("\nEnter a positive integer: ");  
    scanf("%d", &num);
```

```

temp = num;
while (temp > 0) {
    digit = temp % 10;

    // Factorial calculation
    fact = 1;
    for (i = 1; i <= digit; i++) {
        fact = fact * i;
    }

    sum = sum + fact;
    temp = temp / 10;
}

if (sum == num) {
    printf("%d is a Strong Number.\n", num);
} else {
    printf("%d is NOT a Strong Number.\n", num);
}
}

// Function to check Armstrong Number
void checkArmstrong() {
    int num, temp, digit, count = 0, sum = 0;

    printf("\nEnter a positive integer: ");
    scanf("%d", &num);

```

```

temp = num;
// Count number of digits
while (temp > 0) {
    count++;
    temp = temp / 10;
}

temp = num;
while (temp > 0) {
    digit = temp % 10;
    sum = sum + pow(digit, count);
    temp = temp / 10;
}

if (sum == num) {
    printf("%d is an Armstrong Number.\n", num);
} else {
    printf("%d is NOT an Armstrong Number.\n", num);
}
}

// Function to check Harshad Number
void checkHarshad() {
    int num, temp, digit, sum = 0;

    printf("\nEnter a positive integer: ");
    scanf("%d", &num);

```

```

temp = num;
while (temp > 0) {
    digit = temp % 10;
    sum = sum + digit;
    temp = temp / 10;
}

if (num % sum == 0) {
    printf("%d is a Harshad Number.\n", num);
} else {
    printf("%d is NOT a Harshad Number.\n", num);
}
}

// Function to check Duck Number
void checkDuck() {
    int num, temp, digit, isDuck = 0;

    printf("\nEnter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        if (digit == 0) {
            isDuck = 1;
            break;
        }
        temp = temp / 10;
    }
}

```

```
if (isDuck == 1) {  
    printf("%d is a Duck Number.\n", num);  
} else {  
    printf("%d is NOT a Duck Number.\n", num);  
}  
}
```

OUTPUT SCREENSHOT :

```
[manalithakar@Manalis-MacBook-Air ~ % nano 18q1.c
[manalithakar@Manalis-MacBook-Air ~ % gcc 18q1.c -o 18q1
[manalithakar@Manalis-MacBook-Air ~ % ./18q1
```

```
--- MENU ---
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
5. Exit
Enter your choice: 1

Enter a positive integer: 145
145 is a Strong Number.
```

```
--- MENU ---
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
5. Exit
Enter your choice: 2

Enter a positive integer: 371
371 is an Armstrong Number.
```

```
--- MENU ---
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
5. Exit
Enter your choice: 3

Enter a positive integer: 18
18 is a Harshad Number.
```

```
--- MENU ---
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
5. Exit
Enter your choice: 4

Enter a positive integer: 205
205 is a Duck Number.
```

```
--- MENU ---
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
5. Exit
Enter your choice: 5
```

```
Exiting program...
manalithakar@Manalis-MacBook-Air ~ % █
```